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RESEARCH ARTICLE

Proteomic Analysis of the Colistin-resistant *E. coli* Clinical Isolate: Explorations of the Resistome

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Abstract: ***Background*:** Antimicrobial resistance is a worldwide problem after the emergence of colistin resistance since it was the last option left to treat carbapenemase-resistant bacterial infections. The mcr gene and its variants are one of the causes for colistin resistance. Besides mcr genes, some other intrinsic genes are also involved in colistin resistance but still need to be explored.

***Objective*:** The aim of this study was to investigate differential proteins expression of colistin-resistant *E. coli* clinical isolate and to understand their interactive partners as future drug targets.

***Methods*:** In this study, we have employed the whole proteome analysis through LC-MS/MS. The advance proteomics tools were used to find differentially expressed proteins in the colistin-resistant *Escherichia coli* clinical isolate compared to susceptible isolate. Gene ontology and STRING were used for functional annotation and protein-protein interaction networks, respectively.

***Results*:** LC-MS/MS analysis showed overexpression of 47 proteins and underexpression of 74 proteins in colistin-resistant *E. coli*. These proteins belong to DNA replication, transcription and translational process; defense and stress related proteins; proteins of phosphoenol pyruvate phosphotransferase system (PTS) and sugar metabolism. Functional annotation and protein-protein interaction showed translational and cellular metabolic process, sugar metabolism and metabolite interconversion.

***Conclusion*:** We conclude that these protein targets and their pathways might be used to develop novel therapeutics against colistin-resistant infections. These proteins could unveil the mechanism of colistin resistance.

Keywords: Colistin resistance, colistin-resistant *E. coli*, antimicrobial resistance, proteomics, functional annotation, pathways enrichment, STRING.

1. INTRODUCTION

Enterobacterale is a large family of Gram-negative bacteria, including many non-pathogenic symbionts along with pathogens, such as *Escherichia coli, Klebsiella, Enterobacter, Shigella, Proteus, Serratia, and Citrobacter*. Among them, *E.coli* is a major cause of urinary tract infection in the clinical setting, and has been found to be 60-70% resistant to the majority of existing drug families around the globe [1-6]. The emergence and spread of the extended-spectrum cephalosporins and carbapenems resistant *E.coli* threaten the community, worldwide [1-6]. Polymyxin B and colistin (polymyxin E) are used as last resort of drug against the resistant *E.coli* but unfortunately bacteria adopt themechanisms which made it resistant against the colistin, and represent a serious growing threat to public health [7-10].

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Colistin usually disrupts the structure of cell membrane phospholipids, leading to enhanced cell permeability, thereby causing cell death. Colistin resistance is primarily acquired through a diminution of the electrostatic attraction between colistin and the bacterial outer membrane due to the addition of cationic phosphoethanolamine or 4-amino-4-deoxy-L-arabinose (L-Ara4N) moieties to phosphate groups located on the lipid-A component of lipopolysaccharide. Mobilized colistin resistance (MCR) genes (mcr1-10) present on the plasmid are the major cause of colistin resistance in *E. coli* and spread the resistance by means of horizontal gene transfer in the Enterobacteriaceae members [10-19]. Apart from them, a few other chromosomal gene products (mgrB, PhoP-PhoQ and PmrA-PmrB) deregulate the two-component regulatory systems and mutations in these genes leading to loss of function, the potential mechanism responsible for colistin resistance [7-9]. These are a few explanations which demonstrate the mechanisms of colistin resistance, but there is still a significant gap in knowledge with regard to colistin resistance.

However, not much attention has been paid to proteome-based exploration of drug resistance. High-throughput differential expression proteomics and bioinformatics approaches have been used to investigate the novel mechanism of drug resistance in bacteria [20-24]. In this study, we have investigated the differential expressed proteins of colistin-resistant *E. coli* clinical isolate through liquid chromatography coupled with mass spectrometry (LC-MS/MS) approach. We have also conducted an *in-silico* functional analysis of the differentially expressed proteins to discern their role in colistin resistance.

2. MATERIALS AND METHODS

2.1. Ethical Approval

The study was approved by the institutional ethical committee of Aligarh Muslim University and human samples have been collected for bacterial strains characterization. As per the guideline, all subjects were informed of the study procedure and their names have been kept confidential.

2.2. Collection of Bacterial Strains and Characterization

Strains collection was performed from patients attending Jawaharlal Nehru Medical College & Hospital, Aligarh Muslim University. A total of 52 urine samples were collected from infected patients in a wide-mouth sterile container. Isolation was performed by a surface streak procedure on Luria Bertani (LB) broth (Himedia Labs, Mumbai, India) using calibrated loops for semiquantitative method and incubated aerobically at 37°C for 24 hrs, and those cultures which were negative after 24 hrs incubations, were further incubated for 48 hrs. Of 52, 16 samples were characterized to have *E. coli* isolates, confirmed through BD phoenix-100™ Automated Microbiology system using panel NMIC/ID-55 (Gram-negative susceptibility card) and further confirmed by 16s rRNA sequencing using primers, as described previously [25].

2.3. Antimicrobial Susceptibility Testing and Scale-up of the Culture

For the detection of colistin resistance, antimicrobial susceptibility was performed by the standard disc diffusion method using Mueller-Hinton agar (Table **S1**), and minimum inhibitory concentrations (MICs) were determined using the broth microdilution method as per the Clinical and Laboratory Standards Institute Guidelines [26].

2.4. Protein Preparation and Identification by NanoLC-Triple TOF-MS

The protein sample was prepared as per earlier studies [22, 23]; in brief, after washing the cells with normal saline, they were again suspended in the lysis buffer [50 mMTris-HCl containing 10 mM $MgCl_2$, 0.1% sodium azide, 1 mM phenylmethylsulfonyl fluoride (PMSF) and 1 mM ethylene glycol tetraacetic acid (EGTA); pH 7.4] at the concentration of 1g wet weight per 5 ml. Cell lysis was done by intermittent sonication at 35% amplitude (Sonics & Materials Inc, USA) for 10 min at 4°C and the homogenate was centrifuged at 12,000 g for 20 minutes at 4°C to collect the supernatant, which was further precipitated by adding chilled acetone to the supernatant in excess (1:4) and placed at -20°C for overnight. The precipitate was collected by centrifugation (12000 rpm, 20 minutes), washed through acetone and permitted to dry in the suspended buffer for further experiments. The protein concentration was estimated using the Bradford assay [27]. Experiments were performed in technical replicates.

An equal amount of protein was trypsinized and analysed using a Triple TOF 5600 MS (AB-Sciex, USA) equipped with Eskigent MicroLC 200 system (Eskigent, Dublin, CA) with an Eskigent C18-reverse phase column (150 X 0.3 mm, 3 µm, 120Å). In order to generate a spectral library for protein identification experiment, data-dependent analysis (DDA) was performed for the individual protein sample to generate high-quality spectral ion libraries for SWATH analysis with specific parameters. The spectral library was generated using information-dependent acquisition (IDA) mode after injecting tryptic digest of 2 gm on column using Eksigent nanoLC-Ultra™ 2D plus system coupled with SCIEX Triple TOF® 5600 system fitted with NanoSpray III source. The samples were loaded on the trap column (Eksigent ChromXP350 µm x 0.5 mm, 3 µm 120Å) and washed for 30 minutes at 3 µL/minute. A 120 minutes gradient in multiple steps (ranging from 5-50% Acetonitrile in water containing 0.1% formic acid) was set up to elute the peptides from the ChromXP 3-C18, 0.075 x 150 mm, 3µm, 120Å analytical column. Experiments have been repeated in technical replicates.

2.5. Information-dependent Acquisition (IDA)

In IDA, the ion library generation method was used in which a maximum of 20 most intense multiple charged ions per MS cycle were opted to perform MS/MS fragmentation. A dynamic exclusion criterion was implemented to each of the ions for 10 seconds and accumulation time for each MS/MS experiment was set to 70 ms.

2.6. SWATH Parameters for Label-free Quantification

In the SWATH acquisition method, Q1 transmission window was set to 12Da from the mass range for 350-1250Da and a total of 75 windows were acquired independently with an accumulation time of 62ms. Three technical replicates for each of the sets were acquired and total cycle time was kept constant. To generate spectral library, ProteinPilot™ v. 5.0 was used. For label-free quantification, the peak extraction and spectral alignment were performed using PeakView® 2.2 software with specific parameters as follows: the number of peptides as 2, the number of transitions as 5, the peptide confidence as 95%, the XIC width as 30ppm, and the XIC extraction window as 3 min. The data were further subjected to Marker View software V 1.3 (AB Sciex) for statistical data interpretation. In MarkerView, the peak area under the curve for the selected peptides was normalized by internal standard protein (beta galactosidase) spike during the SWATH accumulation. The results were shown as three output files containing AUC of the ions, the summed intensity of peptides for protein and the summed intensity of ions for the peptide. All SWATH acquisition data were processed using the SWATH Acquisition MicroApp 2.0 in PeakView® Software.

2.7. Data Analysis

Data were processed with Protein Pilot Software v. 5.0 (AB SCIEX, Foster City, CA) using the Paragon and Progroup Algorithm. Extended analysis was also performed through integrated tools of Protein Pilot with 1% false discovery rate (FDR).

2.8. Functional Annotation and Pathway Enrichment

Overexpressed and underexpressed proteins data obtained from LC-MS/MS were primarily filtered by excluding redundant and uncharacterized proteins. *E. coli* (strain 0139:H28) was used as a reference strain for functional analysis. Filtered proteins were then aligned against the reference strain proteome at e-value 10^{-6}, considering the first hit of the reference strain protein as a homolog of filtered protein data. For functional annotation of proteins, Gene Ontology (GO) terms associated with the reference strain proteins were used. For functional annotation, a slim version of Gene ontology (GO) was used and GO terms were obtained from Gene Ontology Consortium (http://www.geneontology.org/). PANTHER database was used to perform functional classification of genes obtained from GO analysis [28].

2.9. Protein-protein Interactions (PPIs) Network

To determine the interaction pattern of the up-regulated and down-regulated genes obtained from GO analysis, protein-protein interactions (PPIs) information was obtained from the STRING database v11.0 (http://www.string-db.org/) and visualized in Cytoscape (version 3.8.0). The PPI information has been established by experimental studies or by genomic analyses, like domain fusion, phylogenetical profiling, high-throughput experiments, co-expression studies, and gene neighbourhood analysis that is provided by the STRING database. In the present study, only interactions with a default score of ≥0.4 were considered significant. Proteins showing high interaction were considered as node proteins and their immediate partners were identified and characterized on the basis of function [22-23, 29-31].

3. RESULTS

3.1. Identification of Colistin-resistant *E. coli* Strain and LC-MS/MS-based Proteins Identification

Among 16 *E. coli* isolates identified in this study, 1 isolate was found to be highly resistant to colistin which was further processed for protein sample preparation for proteomics analysis.

We have analyzed the proteome of colistin-resistant clinical bacterial isolate in comparison with the susceptible isolate. Moreover, we have identified the proteome of colistin-resistant isolate by LC-MS/MS and using the SWATH workflow; 1178 proteins were quantified at 1% FDR. Among the differentially expressed proteins, 47 proteins were overexpressed ranging from 47.85 to 2.30 log folds change *vs*. p-value, and 74 proteins were underexpressed ranging from 0.50 to 0.06 log folds change *vs*. p-value, as tabulated in Tables **1** and **2**, respectively. Among them, the majority of proteins belonged to central dogma (replication, transcription and translation-related proteins), proteins that overcome the reactive oxygen species (ROS), proteins related to energy and intermediary metabolism, and proteins of phosphoenol pyruvate phosphotransferase system (PTS). The level of differential expression has been shown as the log folds change *vs*. p-value ratio.

Table 1. Details of the overexpressed proteins in colistin-resistant *Escherichia coli* clinical isolate.

S. No.	Protein Name	Gene	Accession Number	Log Fold Change *vs*. *P*-value
1	30S ribosomal protein S8	rpsH	P0A7W7	47.85
2	50S ribosomal protein L15	rplO	P02413	46.79
3	30S ribosomal protein S3	rpsC	P0A7V3	46.11
4	Phosphate acetyltransferase	eutD	P77218	39.39
5	30S ribosomal protein S13	rpsM	P0A7S9	28.59
6	50S ribosomal protein L14	rplN	P0ADY3	26.35
7	ATP-dependent 6-phosphofructokinase	pfkA	P0A796	24.83
8	30S ribosomal protein S1	rpsA	P0AG67	23.99
9	Acetate kinase	ackA	P0A6A3	23.08
10	Trigger factor	Tig	P0A850	22.27
11	50S ribosomal protein L9	rplI	P0A7R1	20.56
12	Elongation factor Tu	tufA	P0CE47	20.16
13	Esterase ybfF	ybfF	P75736	19.34
14	Phosphoglucosamine mutase	glmM	P31120	17.74

(Table 1) contd….

S. No.	Protein Name	Gene	Accession Number	Log Fold Change *vs. P*-value
15	50S ribosomal protein L19	rplS	P0A7K6	17.53
16	Fructose-bisphosphate aldolase	fbaB	P0A991	17.25
17	Redox-regulated ATPase YchF	ychF	A0A5C9ANY4	16.92
18	50S ribosomal protein L4	rplD	P60723	16.32
19	30S ribosomal protein S20	rpsT	P0A7U7	15.60
20	Molecular chaperone DnaK	dnaK	P0A6Y8	14.47
21	50S ribosomal protein L31 type B	ykgM	P0A7N1	13.04
22	Glycine-tRNA ligase beta subunit	glyS	P00961	12.69
23	DNA gyrase subunit A	gyrA	P0AES4	12.24
24	Bifunctional acetaldehyde-CoA/alcohol dehydrogenase	adhE	A0A444RBE6	11.98
25	30S ribosomal protein S10	rpsJ	P0A7R5	9.53
26	30S ribosomal protein S9	rpsI	P0A7X3	9.47
27	DNA-directed RNA polymerase subunit alpha	rpoA	P0A7Z4	9.27
28	6-phosphogluconate dehydrogenase, decarboxylating	Gnd	P00350	6.99
29	6-phospho-beta-glucosidase	chbF/celF	P17411	6.89
30	Uncharacterized protein (N6_N4_Mtase domain-containing protein)	A1WK_04142	S1H818	6.37
31	Pyruvate dehydrogenase (Acetyl-transferring) E1 component subunit alpha	pdhA	A0A3L0X1S6	6.24
32	DNA-directed RNA polymerase subunit beta	rpoC	P0A8T7	5.67
33	IMP dehydrogenase	guaB	P0ADG7	5.55
34	Sugar ABC transporter ATP-binding protein	FAZ83_08130	A0A4S5AWU0	5.28
35	1,4-dihydroxy-2-naphthoyl-CoA synthase	menB	P0ABU0	5.28
36	Fe-S cluster assembly protein SufB	sufB	P77522	4.94
37	Serine hydroxymethyltransferase	glyA	P0A825	4.49
38	Alkyl hydroperoxide reductase protein C	ahpC	P0AE08	4.16
39	RNA polymerase sigma factor RpoD	rpoD	P00579	4.16
40	Peptide chain release factor 1	prfA	P0A7I0	4.13
41	F0F1 ATP synthase subunit beta	atpF	P0ABA0	4.10
42	Phosphopentomutase	deoB	P0A6K6	3.86
43	Alkyl hydroperoxide reductase protein F	ahpF	P35340	2.56
44	Class 1b ribonucleoside-diphosphate reductase subunit beta	nrdF	A0A6D2W9Y3	2.47
45	Keto-acid formate acetyltransferase	tdcE	P42632	2.42
46	Glutaredoxin 1	grxA	P68688	2.41
47	N-acetylglucosamine-specific PTS system EIICBA components	nagE	A0A0A0H175	2.31

Table 2. Details of the underexpressed proteins in colistin-resistant *Escherichia coli* clinical isolate.

S. No.	Protein Name	Gene	Accession Number	Log Fold Change *vs. P*-value
1	Beta-lactamase	ampC	P00811	0.06
2	Cyclic pyranopterin monophosphate synthase	moaC	P0A738	0.12
3	Uncharacterized protein	-----	W1HG94	0.17
4	CelA subunit of N,N'-diacetylchitobiose PTS permease	celA/chbB	P69795	0.21
5	Possible oxidoreductase	-----	W1HJ45	0.21
6	DNA protection during starvation protein	dps	P0ABT2	0.22
7	Uncharacterized protein/ DUF853 domain-containing protein	-----	W1HJG3	0.23
8	PTS system, trehalose-specific IIB component / IIC component	treB	P36672	0.27
9	Putative aldo/keto reductase	iolS_3	A0A166BDB1	0.27
10	Outer membrane protein assembly factor BamD	bamD	P0AC02	0.28
11	Aldehyde-alcohol dehydrogenase	-----	W1HGN5	0.28
12	ATP-dependent Clp protease ATP-binding subunit ClpA	clpA	P0ABH9	0.28
13	Succinate-semialdehyde dehydrogenase [NADP+]	gabD	P25526	0.28

(Table 2) contd….

S. No.	Protein Name	Gene	Accession Number	Log Fold Change *vs. P*-value
14	Bacterioferritin	bfr	P0ABD3	0.29
15	Osmotically inducible protein OsmY	osmY	P0AFH8	0.29
16	Serine protein kinase (PrkA protein), P-loop containing	yeaG	C3T6U2	0.29
17	Periplasmic serine endoprotease DegP-like	degP	A0A4S4PSC0	0.30
18	Probable secreted protein	-----	W1H7R9	0.31
19	Osmotically inducible lipoprotein E	osmE	P0ADB2	0.32
20	PTS system fructose-specific transporter subunit IIBC	fruA	P20966	0.33
21	Phosphoenolpyruvate carboxykinase (ATP)	pckA	P22259	0.33
22	Aminomethyltransferase	gcvT	P27248	0.34
23	Alcohol dehydrogenase	adhP	P39451	0.34
24	Sigma-70, region 3	EC970259_3213	V6FSH2	0.35
25	Ribitol operon repressor protein	UMNK88_2636	A0A0E0U0A6	0.35
26	UPF0325 protein YaeH	yaeH	P62768	0.35
27	PTS enzyme IIC, mannose-specific	manY	P69801	0.35
28	UPF0098 protein YbhB	ybhB	P12994	0.36
29	Ribosome hibernation protein YfiA	raiA	P0AD49	0.37
30	YcfF/hinT protein: a purine nucleoside phosphoramidase	hinT	C3TDT2	0.37
31	Uncharacterized membrane protein YqjD	yqjD	P64581	0.37
32	Major outer membrane lipoprotein	lpp	P69776	0.37
33	Transaldolase	talA	P0A867	0.37
34	PTS system, cellobiose-specific IIB component	gmuB	A0A0J3VF43	0.37
35	Glycine dehydrogenase (decarboxylating)	gcvP	P33195	0.37
36	Uncharacterized protein YoaC	yoaC	P64490	0.37
37	Inorganic pyrophosphatase	ppa	P0A7A9	0.37
38	Putative carbon starvation protein	yjiY	A0A0G9FL58	0.38
39	UPF0337 protein yjbJ	yjbJ	P68206	0.38
40	Transcriptional regulatory protein YciT	yciT	Q7UCQ7	0.39
41	Glutamate dehydrogenase	-----	W1HFF8	0.39
42	Catalase	katE	P21179	0.39
43	Uncharacterized protein YaiA	yaiA	P0AAN5	0.40
44	GTPase Obg	obgE	P42641	0.40
45	Outer membrane lipoprotein pcp	pcp	H4LC23	0.40
46	Glutathione S-transferase	gstA	P0A9D2	0.40
47	Glucose-1-phosphate adenylyltransferase	glgC	P0A6V1	0.40
48	Universal stress protein G	uspG	P39177	0.41
49	AAC(6')-Ib-cr (6'-N-aminoglycoside acetyltransferase aac(6')-Ib)	aac(6')-Ib	D9MWQ6	0.41
50	Fructose-specific phosphocarrier protein HPr/PTS system, fructose-specific IIA component	fruB	A0A023Z0Q4	0.42
51	D-serine dehydratase	dsdA	P00926	0.43
52	Beta-galactosidase	lacZ	P00722	0.43
53	Acetate operon repressor	iclR	P16528	0.43
54	Small HspC2 heat shock protein	----	W1HLV2	0.44
55	Universal stress protein F	----	W1H8L1	0.44
56	Mannose-specific PTS system EIID component	manZ	P69805	0.44
57	Carbon storage regulator	csrA	P69913	0.45
58	Cell division protein ZapE	zapE/yhcM	P64612	0.45
59	PTS system, mannose-specific IIA component / IIB component	manX	P69797	0.45
60	Glycerol kinase	glpK	P0A6F3	0.46
61	Autonomous glycyl radical cofactor	grcA	P68066	0.46
62	Uronate isomerise	uxaC	P0A8G3	0.47
63	PLP-dependent aminotransferase YfdZ	alaC	P77434	0.47

(Table 2) contd….

S. No.	Protein Name	Gene	Accession Number	Log Fold Change *vs. P*-value
64	Cobalt-precorrin-4 C11-methyltransferase	-----	W1HJ18	0.47
65	UPF0441 protein YgiB	ygiB	P0ADT2	0.48
66	UPF0597 protein YhaM	yhaM	P42626	0.48
67	Glycerol dehydrogenase	gldA	P0A9S5	0.49
68	Arginine ABC transporter, periplasmic arginine-binding protein ArtI	artI	P30859	0.49
69	PTS sugar transporter	ptsH	P0AA04	0.49
70	Periplasmic serine endoprotease DegP-like	degP	P0C0V0	0.50
71	Class B acid phosphatise	aphA	P0AE22	0.50
72	Malate dehydrogenase	mdh	P61889	0.50
73	Periplasmic protein	spy	P77754	0.50
74	Curved DNA-binding protein	cbpA	P36659	0.50

Table 3. Functional analysis of up-regulated genes using GO analysis related to *Escherichia coli* (strain 0139:H28).

GO ID	Function	Gene Name	No. of Genes in which GO Term was Found
(A) Biological Function			
GO:0005975	Carbohydrate metabolic process	ascB,eno,gapC,gatY,glmM,mdh,pfkA,pflB2,pgl,pykF	10
GO:0006091	Generation of precursor metabolites and energy	eno,mdh,pfkA,pgl,pykF	5
GO:0006259	DNA metabolic process	EcE24377A_4904,gyrA,recA	3
GO:0006399	tRNA metabolic process	argS,gltX,glyQ,tgt	4
GO:0006412	Translation	argS,fusA,gltX,glyQ,prfA,rplB,rplD,rplN,rplO,rplS,rpmC, rpmE1,rpsB,rpsC,rpsD,rpsE,rpsG,rpsH,rpsI,rpsJ,rpsL,rpsM,rpsT,tuf1	24
GO:0006457	Protein folding	dnaK,groL	2
GO:0006520	Cellular amino acid metabolic process	EcE24377A_3199,argI,argS,gltX,glyA,glyQ	6
GO:0006605	Protein targeting	secA	1
GO:0006790	Sulfur compound metabolic process	EcE24377A_2591,ackA,sufB	3
GO:0006810	Transport	atpA,atpD,ptsI	3
GO:0006950	Response to stress	ahpC,recA,sodA	3
GO:0008150	Biological process	EcE24377A_3199,EcE24377A_4904,ackA,adhE,ahpC,ascB,atpA,atpD, deoB,gapC,glmM,guaB,hflB,mdh,pfkA,ptsI,pykF,rpoD,sodA	19
GO:0009056	Catabolic process	eno,gatY,hflB,pfkA,pykF	5
GO:0009058	Biosynthetic process	EcE24377A_2591,ackA,argI,atpA,atpD,deoB, glyA,guaB,gyrA,menB,rpoA,rpoB,rpoC,rpoD,sufB,tgt	16
GO:0015031	Protein transport	secA	1
GO:0022607	Cellular component assembly	sufB	1
GO:0034641	Cellular nitrogen compound metabolic process	EcE24377A_2591,ackA,atpA,atpD,deoB,eno, glyA,guaB,pfkA,pykF,rpoA,rpoB,rpoC,rpoD,tgt	15
GO:0034655	Nucleobase-containing compound catabolic process	deoB	1
GO:0042592	Homeostatic process	ahpC	1
GO:0044281	Small molecule metabolic process	EcE24377A_2591,ackA,adhE,atpA,atpD, deoB,eno,gapC,gatY,glyA,guaB,mdh,menB,pfkA,pgl,pykF,tgt	17
GO:0051186	Cofactor metabolic process	EcE24377A_2591,ackA,glyA,menB,sufB	5
GO:0051276	Chromosome organization	gyrA,hupB	2
GO:0055085	Transmembrane transport	atpA,atpD,secA	3
(B) Cellular Component			
GO:0005575	Cellular component	EcE24377A_2592,atpA,atpD,eno,hflB,secA	6
GO:0005576	Extracellular region	eno	1
GO:0005622	Intracellular	atpA,atpD	2
GO:0005623	Cell	ahpC	1
GO:0005694	Chromosome	gyrA	1
GO:0005737	Cytoplasm	EcE24377A_2591,ackA,ahpC,argI,argS,deoB,eno,fusA,gltX,glyA, glyQ,groL,gyrA,pfkA,pflB2,prfA,ptsI,recA,rpoD,secA,tuf1	21
GO:0005829	Cytosol	eno	1

(Table 3) contd….

GO ID	Function	Gene Name	No. of Genes in which GO Term was Found
(A) Biological Function			
GO:0005840	Ribosome	rplB,rplD,rplN,rplO,rplS,rpmC,rpmE1,rpsB, rpsC,rpsD,rpsE,rpsG,rpsH,rpsI,rpsJ,rpsL,rpsM,rpsT	18
GO:0005886	Plasma membrane	atpA,atpD,hflB,secA	4
GO:0032991	Protein-containing complex	atpA,atpD,eno,rplB,rplN,rplO,rpsB,rpsD,rpsE,rpsG,rpsL	11
(C) Molecular Function			
GO:0003674	Molecular function	EcE24377A_0714,EcE24377A_1529,EcE24377A_2591, EcE24377A_3199,EcE24377A_4904,ackA,ahpC,argI, argS, ascB,atpA,atpD,deoB,dnaK,engD,fusA,gapC, gatY,gltX,glyA,glyQ, groL,guaB,gyrA,hflB, mdh,menB,pfkA,pflB2,pgl,ptsI,pykF,recA, rplB,rpoA, rpoB,rpoC,rpoD,rpsC,rpsM,secA,tgt,tuf1	43
GO:0003677	DNA binding	EcE24377A_4904,gyrA,hupB,recA,rpoA,rpoB,rpoC,rpoD	8
GO:0003700	DNA-binding transcription factor activity	rpoD	1
GO:0003723	RNA binding	gltX,rplB,rplD,rplN,rplO,rpsC,rpsD, rpsE,rpsG,rpsH,rpsJ,rpsL,rpsM,rpsT	14
GO:0003729	mRNA binding	rpsC	1
GO:0003735	Structural constituent of ribosome	rplB,rplD,rplN,rplO,rplS,rpmC,rpmE1,rpsB,rpsC, rpsD,rpsE,rpsG,rpsH,rpsI,rpsJ,rpsL,rpsM,rpsT	18
GO:0003924	GTPase activity	fusA,tuf1	2
GO:0008135	Translation factor activity, RNA binding	fusA,prfA,tuf1	3
GO:0008168	Methyltransferase activity	EcE24377A_4904	1
GO:0008233	Peptidase activity	hflB	1
GO:0016301	Kinase activity	EcE24377A_3199,ackA,pfkA,ptsI,pykF	5
GO:0016491	Oxidoreductase activity	adhE,ahpC,gapC,guaB,mdh,sodA	6
GO:0016746	Transferase activity, transferring acyl groups	EcE24377A_2591,pflB2	2
GO:0016757	Transferase activity, transferring glycosyl groups	tgt	1
GO:0016779	Nucleotidyltransferase activity	rpoA,rpoB,rpoC	3
GO:0016798	Hydrolase activity, acting on glycosyl bonds	ascB	1
GO:0016829	Lyase activity	eno,gatY,menB	3
GO:0016853	Isomerase activity	deoB,glmM,gyrA	3
GO:0016874	Ligase activity	argS,gltX,glyQ	3
GO:0016887	ATPase activity	EcE24377A_1529,engD,gyrA,hflB,recA	5
GO:0019843	rRNA binding	rplB,rplD,rplN,rplO,rpsC,rpsD,rpsE,rpsG,rpsH,rpsL,rpsM,rpsT	12
GO:0022857	Transmembrane transporter activity	atpA,atpD	2
GO:0043167	Ion binding	EcE24377A_1529,ackA,adhE,argI, argS,atpA,atpD,deoB,dnaK,engD,eno, fusA,gatY,glmM,gltX,glyA,glyQ,groL,guaB,gyrA,hflB,pfkA, ptsI,pykF,recA,rpoC,secA,sodA,tgt,tuf1	30
GO:0051082	Unfolded protein binding	dnaK,groL	2

3.2. Functional Annotation and Pathway Analyses

The overexpressed and underexpressed proteins obtained from LC-MS/MS data were enriched using *E. coli* (strain 0139:H28) as a control strain to perform all functional annotation studies, and the results are tabulated in Tables **3** and **4**. Based on the PANTHER database, GO analysis of functional enrichment revealed that the overexpressed proteins of colistin-resistant isolate belonged to the class of translational and cellular carbohydrate, amino acid and nitrogen compound metabolic process (Figure **1**), whereas the underexpressed proteins belonged to the class of metabolite interconversion enzymes, transcriptional regulator and nucleic acid-binding proteins (Figure **2**).

3.3. Protein-Protein Interactions Network (PPIs)

PPIs networks were constructed by mapping the overexpressed and underexpressed proteins against *E. coli* (strain 0157H7) using the STRING database, as represented in Figures **3** and **4**, respectively. The interactome of over-expressed proteins (color-coded) showed a strong interaction network in ribosomal 50S (pink) & 30S (green), chaperones (cyan), ATP synthesis (blue) and DNA binding proteins (yellow), as represented in Figure **3**. The interactome of underexpressed proteins (color-coded) showed a good interaction network in citrate synthase (yellow) and PTS proteins (green), as represented in Figure **4**.

Table 4. Functional analysis of down-regulated genes using GO analysis related to *Escherichia coli* (strain 0139:H28).

GO ID	Function	Gene Name	No. of Genes in which GO Term was Found
(A) Biological Function			
GO:0005975	Carbohydrate metabolic process	csrA,glgC,glpK,lacZ,mdh,pckA,treC,uxaC	8
GO:0006091	Generation of precursor metabolites and energy	glgC,mdh	2
GO:0006412	Translation	csrA	1
GO:0006457	Protein folding	cbpA	1
GO:0006464	Cellular protein modification process	EcE24377A_2007	1
GO:0006520	Cellular amino acid metabolic process	dsdA,gcvT	2
GO:0006810	Transport	EcE24377A_1739,artI,bfr,fruB,manX,manY,ptsH,treB	8
GO:0006950	Response to stress	katE	1
GO:0008150	Biological process	EcE24377A_1122,EcE24377A_1832,EcE24377A_2007, EcE24377A_2882,adhE,ampC,bfr,clpA,cysG, dps,fruB,glgC,glpD,glpK,katE,lacZ, maeB,mdh,ppa,rpoS,treB,treC	22
GO:0009056	Catabolic process	ampC,gcvT,glpK,katE,treC,uxaC	6
GO:0009058	Biosynthetic process	EcE24377A_1487,EcE24377A_2668,cysG,glgC,iclR,malI,moaC,pck-A,rpoS	9
GO:0034641	Cellular nitrogen compound metabolic process	EcE24377A_1487,cysG,iclR,malI,rpoS	5
GO:0034655	Nucleobase-containing compound catabolic process	csrA	1
GO:0042254	Ribosome biogenesis	obg	1
GO:0042592	Homeostatic process	bfr,dps	2
GO:0044281	Small-molecule metabolic process	adhE,cysG,glpK,maeB,mdh,pckA,uxaC	7
GO:0051186	Cofactor metabolic process	cysG,katE,moaC	3
GO:0051276	Chromosome organization	dps	1
GO:0055085	Transmembrane transport	EcE24377A_1739,artI,manX,ptsH,treB	5
(B) Cellular Component			
GO:0005575	Cellular component	EcE24377A_3571,ampC,artI,cbpA,dps,lpp, manX,manY,osmE,slyB,spy,treB	12
GO:0005623	Cell	bfr,dps	2
GO:0005737	Cytoplasm	cbpA,csrA,dps,fruB,glpD,manX,obg,pckA,ppa,ptsH,rpoS,treC	12
GO:0005886	Plasma membrane	treB	1
GO:0032991	Protein-containing complex	glpD,lacZ	2
(C) Molecular Function			
GO:0003674	Molecular function	EcE24377A_1122,EcE24377A_1739,EcE24377A_1845, EcE24377A_2668,EcE24377A_3456,EcE24377A_3571,acnA,ampC, clpA,cysG,fruB,gcvT,glgC,glpK,grcA,hinT,katE,lacZ,maeB,manX, mdh,obg,pckA,ppa,ptsH,rpoS,treB,treC,yihX	29
GO:0003677	DNA binding	EcE24377A_1487,cbpA,dps,iclR,malI,rpoS	6
GO:0003700	DNA-binding transcription factor activity	EcE24377A_1487,rpoS	2
GO:0003723	RNA binding	csrA	1
GO:0003729	mRNA binding	csrA	1
GO:0003924	GTPase activity	obg	1
GO:0008168	Methyltransferase activity	cysG,gcvT	2
GO:0008233	Peptidase activity	clpA	1
GO:0016301	Kinase activity	EcE24377A_2007,fruB,glpK,treB	4
GO:0016491	Oxidoreductase activity	EcE24377A_1122,EcE24377A_1832,adhE,bfr,cysG, dps,glpD,katE,maeB,mdh	10
GO:0016746	Transferase activity, transferring acyl groups	maeB	1

(Table 4) contd….

GO ID	Function	Gene Name	No. of Genes in which GO Term was Found
GO:0016765	Transferase activity, transferring alkyl or aryl (other than methyl) groups	EcE24377A_1845,EcE24377A_3456	2
GO:0016779	Nucleotidyltransferase activity	glgC	1
GO:0016798	Hydrolase activity, acting on glycosyl bonds	lacZ,treC	2
GO:0016810	Hydrolase activity, acting on carbon-nitrogen (but not peptide) bonds	ampC	1
GO:0016829	Lyase activity	acnA,cysG,dsdA,maeB,moaC,pckA	6
GO:0016853	Isomerase activity	uxaC	1
GO:0016887	ATPase activity	clpA	1
GO:0022857	Transmembrane transporter activity	EcE24377A_1739,artI,manX,ptsH,treB	5
GO:0043167	Ion binding	EcE24377A_1122,EcE24377A_2668,adhE,bfr,clpA,dps, dsdA,glgC,glpK,katE,lacZ,maeB,obg,pckA,ppa	14
GO:0051082	Unfolded protein binding	cbpA	1

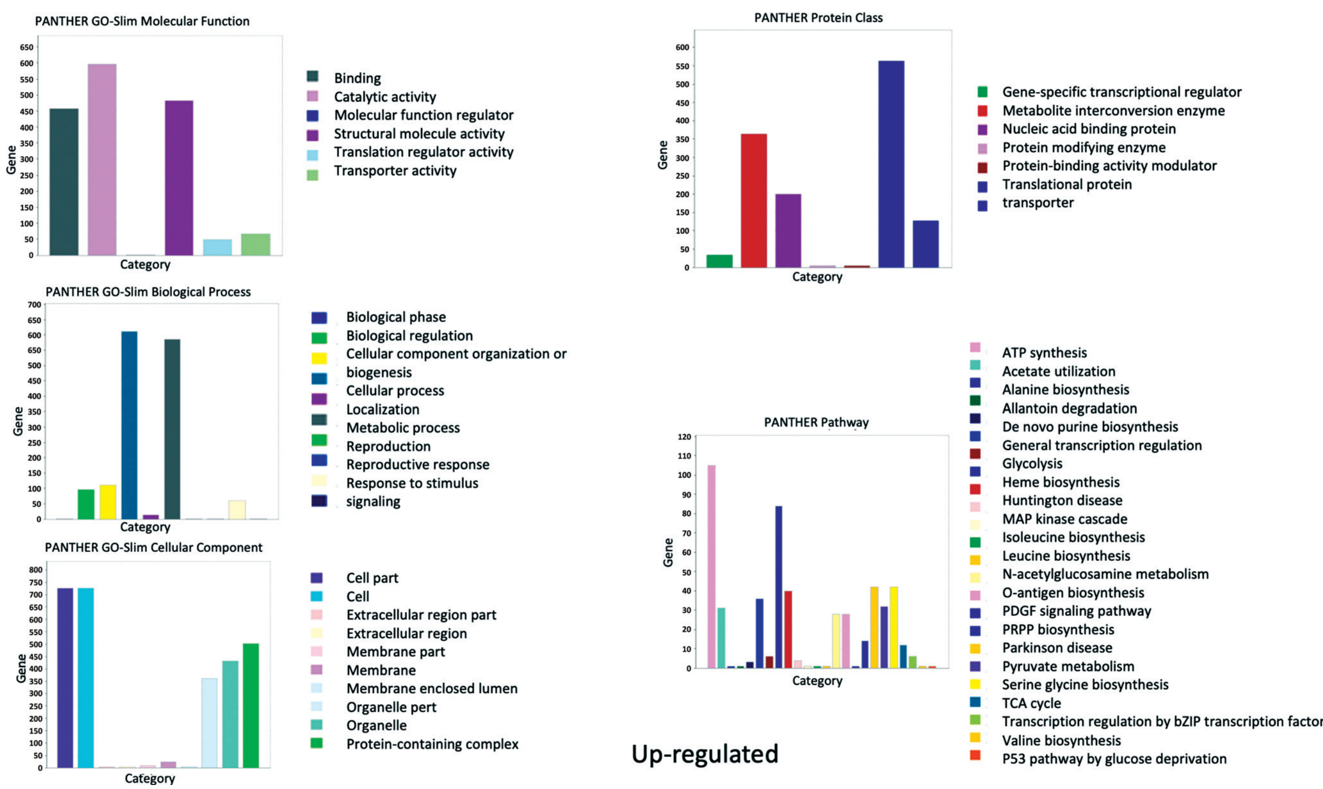

Figure 1. The overexpressed protein lists were submitted to the 'Compare gene list' tool in the PANTHER database to gain insights on the gene ontology biological processes, molecular function and cellular component present in the list in order to highlight protein functions and class. Overexpressed proteins involve a pool of translational, energy metabolism and respiration related proteins that protect from ROS. (*A higher resolution / colour version of this figure is available in the electronic copy of the article*).

4. DISCUSSION

The rapid spread of colistin-resistant *E. coli* has become a great concern for physicians and healthcare workers across the world. The presence of mobilized colistin resistance (MCR) genes (mcr1-10) on the plasmid is the main mechanism of action of colistin resistance in *E. coli* [10-19]. Apart from that, a few other chromosomal gene products that deregulate the two-component regulatory systems (phoPQ and pmrAB), and their negative regulator (mgrB) that leads to the alteration of the lipid-A moieties, also exhibit colistin resistance [7-10]. Mutations in genes of two-component regulatory systems promote the lipid-A modification and contribute to colistin resistance. As these mechanisms are not completely elucidated, the complete resistome is still at the fragmentary stage. In this study, we have observed the differential expressions of a panel of prot-

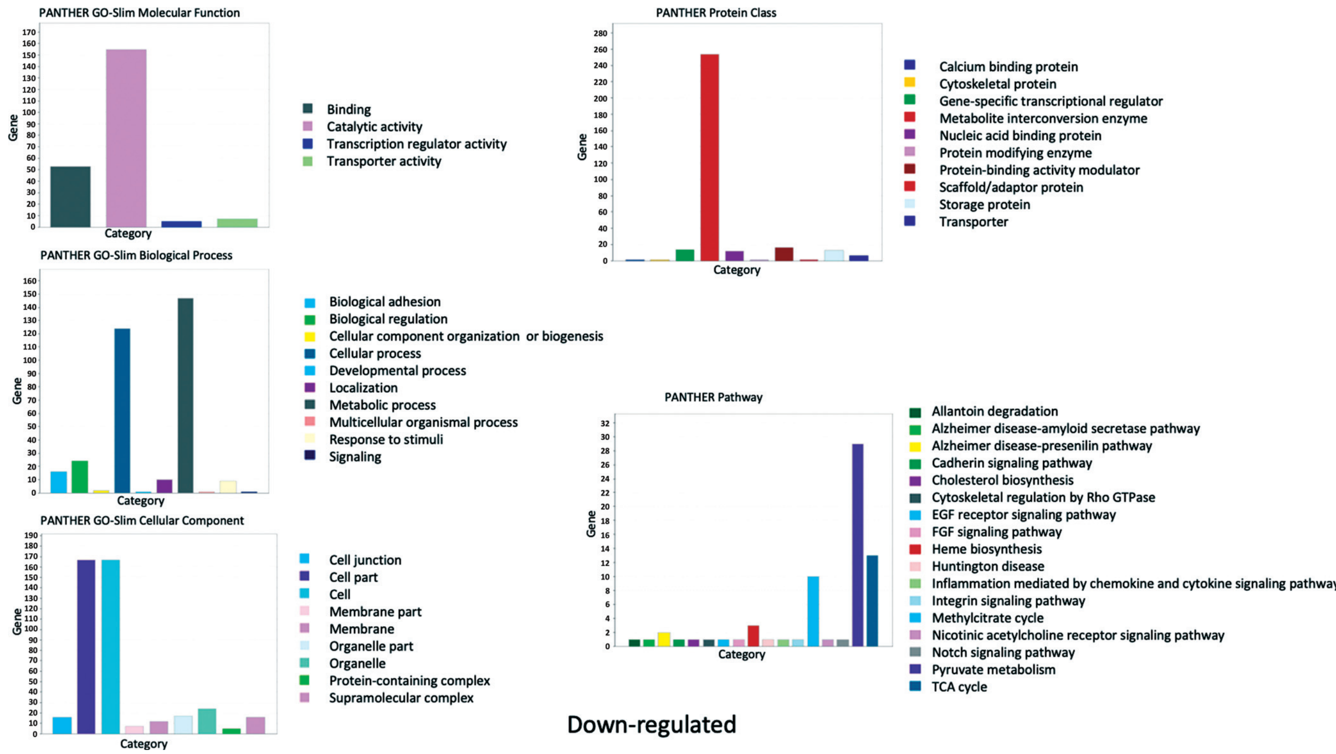

Figure 2. The underexpressed protein lists were submitted to the 'Compare gene list' tool in the PANTHER database to gain insights on the Gene Ontology Biological processes, Molecular function and Cellular component present in the list to highlight protein functions. The underexpressed proteins are mostly related to PTS system, and are metabolite inter-conversion enzymes like transferases. (*A higher resolution / colour version of this figure is available in the electronic copy of the article*).

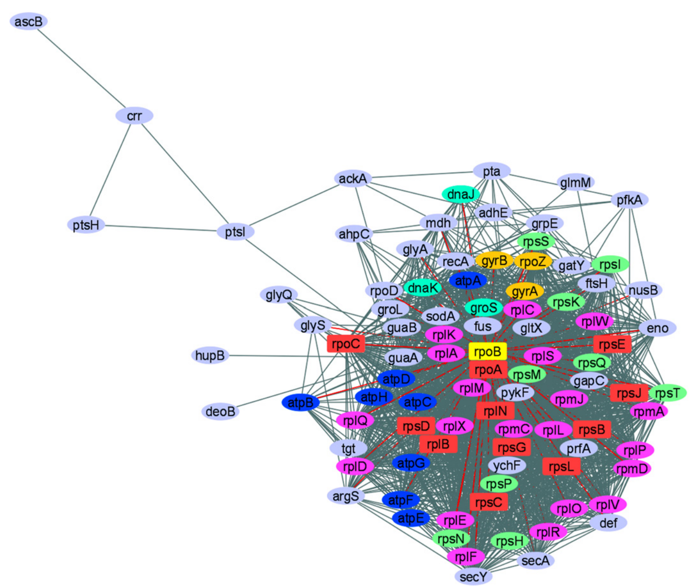

Figure 3. Interaction networks for overexpressed genes of *E. coli* using *E. coli* O157H7 homologs. Interacting proteins were color-coded by functions, ATP synthase subunit proteins like atpB, atpC, atpH, atpE, atpG, atpF, atpA, and atpD (blue), 50S ribosomal subunit proteins like rplF, rplX, rplD, rplE, rplL, rpmJ, rpmC, rpmA, rplV, rplW, rplS, rplR, rpmD, rplC, rplP, rplA, rplK, rplO, rplQ, and rplM (pink), 30S ribosomal subunit proteins like rpsN, rpsS, rpsQ, rpsT, rpsP, rpsH, rpsI, and rpsKrpsM (green), chaperone proteins like groS, dnaJ, and dnaK (cyan), and DNA binding proteins like rpoZ, gyrB, and gyrA (mustard yellow). Square nodes in red color and yellow colour indicate overexpressed proteins like rpsE, rplB, rpoA, rpsD, rpsC, rpsG, rpsB, rpsL, rplN, rpsJ, and rpoC, and top-most node proteins like rpoB, respectively, mapped on *E. coli*. (*A higher resolution / colour version of this figure is available in the electronic copy of the article*).

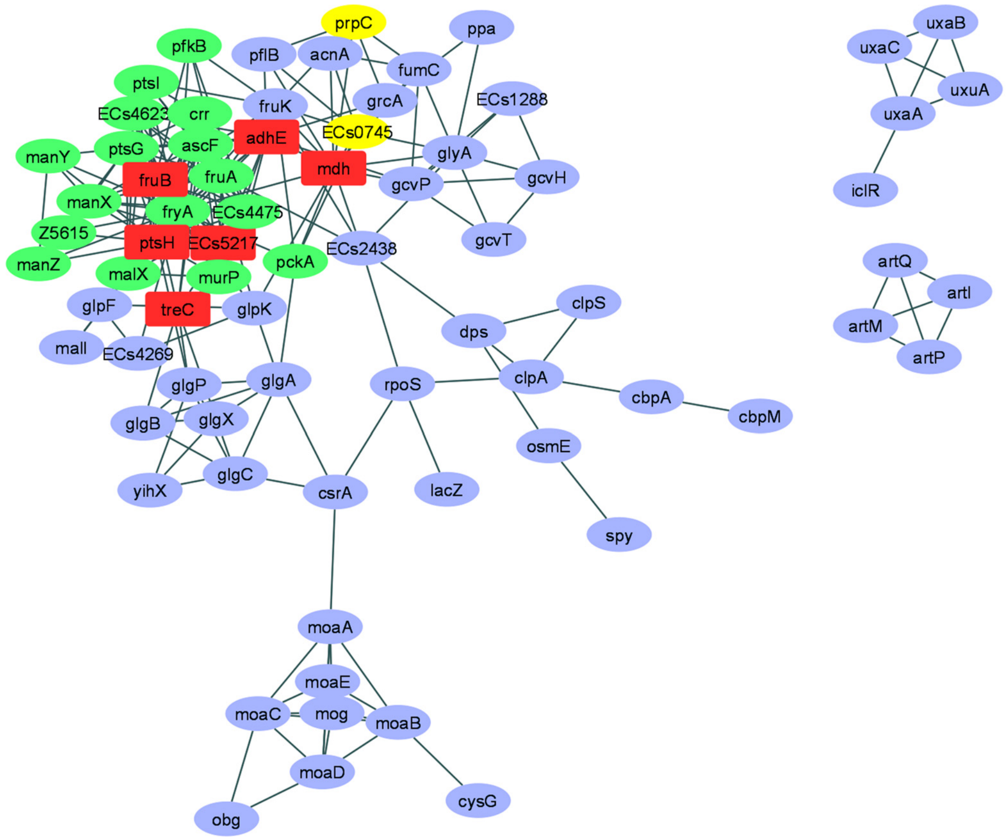

Figure 4. Interaction networks for underexpressed genes of *E. coli* using *E. coli* O157H7 homologs. Interacting proteins were color-coded by functions, phosphotransferase system (PTS) proteins like manY, Z5615, manZ, manX, ptsG, murP, ascF, fryA, malX, ptsI, ECs4475, fruA, pfkB, crr, ECs4623, and pckA(green) and citrate synthase proteins like prpC and ECs0745 (yellow). Square nodes in red color indicate underexpressed proteins like fruB, ptsH, ECs5217, treC, adhE, and mdh mapped on *E. coli*. (*A higher resolution / colour version of this figure is available in the electronic copy of the article*).

eins (47 overexpressed and 74 underexpressed proteins) in colistin-resistant *E. coli*, which could contribute to colistin resistance.

4.1. Overexpression of Replication and Trans- cription Related Proteins

Data of this study show that major proteins related to DNA replication as well as transcription are overexpressed in colistin-resistant clinical isolates. These proteins are DNA gyrase subunit A, DNA-directed RNA polymerase subunit alpha, DNA-directed RNA polymerase subunit beta and RNA polymerase sigma factor RpoD. The overexpression of these universal enzymes leads to the replication and transcription of various genes that are further translated to their protein products, which could be potentially involved in drug resistance. Thus, the overexpression of these universal enzymes of central dogma could be indirectly related to drug resistance.

4.2. Overexpression of Proteins of Ribosomal Machinery Complex and Translational Process

A cluster of proteins related to ribosomal machinery complex and translational process have been found to be overexpressed in colistin-resistant *E. coli* clinical isolate as compared to susceptible isolate. These proteins are Elongation factor Tu, Peptide chain release factor 1, Glycine-tRNA ligase beta subunit, 30S ribosomal protein S1, 30S ribosomal protein S3, 30S ribosomal protein S8, 30S ribosomal protein S9, 30S ribosomal protein S10, 30S ribosomal protein S13, 30S ribosomal protein S20, 50S ribosomal protein L4, 50S ribosomal protein L9, 50S ribosomal protein L14, 50S ribosomal protein L15, 50S ribosomal protein L19 and 50S ribosomal protein L31B. Elongation factor Tu (EF-Tu) is a GTP-associated protein that lengthens the peptide chain in the translational process [32, 33]. Peptide chain release factor 1 (RF1) is a class I release factor (similar to RF2) which is involved in polypeptide chain termination event (a crucial step for recognizing the UAG and UAA termination codons) [33-35]. RF3 is a class II release factor with GTPase activity that accelerates the activity of RF1 and shows functional similarity to the EF-G [36]. The rest of the 30S and 50S ribosomal proteins make a complete ribosomal protein translational machinery complex, which is active functionally and synthesizes the nascent polypeptides. Interactome analysis of this cluster of proteins exhibited their close interactions with other proteins in the translational process. In our earlier studies, we have reported the overexpressions of translational process-related proteins in the drug-resistant *Klebsiella pneumoniae* and *Mycobacterium tuberculosis* clinical isolates [21-22, 37], and demonstrated their link to drug resistance. Therefore, we suggest that the overexpression of this cluster of proteins may speed up the proteins biosynthesis in colistin-resistant *E. coli* clinical isolates, and could be explored as the potential mechanism or target for colistin resistance.

4.3. Overexpression of Proteins Involved in Protection from ROS and other Stress Factors

Proteins that protect the bacteria from different stress factors are overexpressed in colistin-resistant *E. coli* clinical isolates. These are alkyl hydroperoxide reductase protein C (AhpC), alkyl hydroperoxide reductase protein F (AhpF), Glutaredoxin 1 (grx1), molecular chaperone DnaK and trigger factor. AhpC, AhpF and Grx1 belong to a class of peroxiredoxins and offer antioxidant defenses to microbes against ROS (hydrogen peroxide and organic hydroperoxides), which catalytically reduce the peroxides by redox-active cysteines [38]. By using NADPH or NADH as a reducing agent, AhpC and AhpF convert the alkyl hydroperoxides to the corresponding alcohols and protect the bacteria from oxidative damage [38-40]. We have previously reported AhpC and other peroxidases to be overexpressed in carbapenem-resistant *Klebsiella pneumoniae* clinical isolate under meropenem pressure [37]. Molecular chaperone DnaK and trigger factor act as a chaperone and are involved in various functions. DnaK is involved in DNA replication mediating interaction with other proteins as well as stress tolerance, whereas the trigger factor maintains the open conformation of newly synthesized secretory as well as non-secretory proteins and stress tolerance, thus preventing aggregation [41, 42]. We earlier suggested that the trigger factor through interaction with ferritin and bacterioferritin might contribute to resistance [30]. Cumulatively, this group of proteins acts as a bacterial scavenging or defense protein, which protects the bacteria from various stress factors; therefore, the overexpression of these proteins might protect drug-resistant *E. coli* from drug stress. The novel mechanism of drug resistance needs to be explored further, which could open the novel avenue for future research.

4.4. Underexpression of the Phosphotransferase System (PTS) and Associated Regulatory Proteins Involved in Sugar Transport and Metabolism

In this study, we have identified a panel of proteins belonging to PTS and its associated proteins. These proteins are CelA subunit of N,N'-diacetylchitobiose PTS permease, PTS system trehalose-specific IIB component/IIC component, PTS system fructose-specific transporter subunit IIBC, PTS enzyme IIC, mannose-specific, PTS system cellobiose-specific IIB component, Fructose-specific phospho carrier protein HPr/PTS system fructose-specific IIA component, Mannose-specific PTS system EIID component, PTS system mannose-specific IIA component/IIB component, PTS sugar transporter, Glycerol kinase, Phosphoenolpyruvate carboxykinase, Carbon storage regulator, Beta-galactosidase, Ribitol operon repressor protein, and Acetate operon repressor. It has been reported that proteins of PTS are involved in the transport of several glucose-related carbon sources and N-Acetyl-D-glucosamine (NAG), which is a major component of lipopolysaccharide present in the outer membrane of cell [43, 44]. Researchers suggest that proteins of PTS can be a part of adaptive colistin resistance in *E. coli* because colistin pressure induces the expression of resistance-inducing proteins, while in the absence of colistin, the expression decreases rapidly [45]. Therefore, this study indicates that PTS might be an important component of the intrinsic resistome that confers colistin resistance. PTS is a complex system comprising a bunch of proteins that are involved in carbohydrate transport as well as phosphorylation (linked processes), and regulate the carbon metabolism in bacteria [46]. Cyclic-AMP receptor protein (Crp) is the major regulator in enteric bacteria, which regulates the expression of various operons/genes [47, 48]. PTS tightly regulates the concentration of Crp/cAMP complex, which generates various active state conformations that subsequently regulate the expression of genes involved in major vital processes [49, 50]. A few earlier published researches have shown Man-PTS expression to be decreased in bacteriocins resistant strains of *Listeria lectis* and *Enterococcus faecalis* as compared to the wild type strains [51, 52]. A study also showed that proteins related to the PTS system sorbitol transporters were decreased in gentamycin-resistant *Lactobacillus casei* [53]. Similarly, in the present study, the proteins related to the PTS system showed decreased expressions, which may cumulatively contribute to colistin resistance. Therefore, proteins of the PTS system, its pathways and the interactive partners may be explored as potential drug targets, which could be further exploited for novel drug discovery against colistin-resistant infections.

In this study, we also employed GO and STRING analysis for functional enrichment of proteins and interactome prediction, respectively, through MS-based identified proteins. *Escherichia coli* (strain 0139:H28) strain was used as a reference strain and overexpressed proteins were found to belong to a class of translational and cellular carbohydrate, amino acid and nitrogen compound metabolic process (Figure **1**; Table **3**), while underexpressed proteins belonged to a class of metabolite interconversion enzymes, transcriptional regulator and nucleic acid-binding proteins (Figure **2**; Table **4**). Here, we hypothesized that the overexpressed proteins are mostly affected by the cellular defense, transcription and translational pathways, which could potentially be general stress factors activated during the drug resistance. The interactome of the overexpressed proteins showed to possess chaperone, energy metabolism, DNA replication, transcription and translational related proteins, while the interactome of underexpressed proteins showed to comprise sugar metabolism and phosphoenolpyruvate phosphotransferase system (PTS) related proteins, which could reveal the new mechanism of colistin resistance (Figures **3** and **4**) and could be exploited as a novel drug target for drug discovery. This study shows that the cumulative effect of over and underexpressed proteins and their respective pathways might be helpful in elucidating the complete resistome of *E. coli*, which has not been earlier explored.

CONCLUSION

In this study, we have employed the discovery proteomics coupled with bioinformatic approaches and found a few groups of differentially expressed proteins in colistin-resistant *E. coli* clinical isolate. These proteins belong to DNA replication, transcription and translational process; defense and stress related proteins; proteins of phosphotransferase system (PTS) and sugar metabolism, and indicate their direct or indirect involvement in colistin resistance. We suggest that these differentially expressed proteins, their interactive protein partners and pathways could cumulatively open the new vistas to further understand complete colistin resistance in gram-negative bacterium like *E. coli*. Further research on these protein targets and their pathways might be employed for the development of novel therapeutics against colistin-resistant infections, so that the situation of the emergence of antibiotic-resistant pathogenic strains could be managed.

LIST OF ABBREVIATIONS

MIC = Minimum Inhibitory Concentration

ESBLs = Extended Spectrum Beta-lactamases

CLSI = Clinical and Laboratory Standards Institute

STRING = Search Tool for the Retrieval of Interacting Genes/Proteins

LB = Luria-Bertani

AUTHORS' CONTRIBUTION

DS designed the concept of the study, conducted the experiment and prepared the draft of the manuscript. NA identified and characterized colistin-resistant strain. AUK designed the study concept and checked the final draft of the manuscript for submission. MA and MK performed *in-silico* analysis and contributed to the writing of the manuscript.

ETHICS APPROVAL AND CONSENT TO PARTICIPATE

The study was approved by the institutional ethical committee of Aligarh Muslim University, and human samples have been collected for bacterial strains characterization. Approval no.: 151/201517/PDFWM-2015-2017-UTT-31140 (SAII).

HUMAN AND ANIMAL RIGHTS

No animals were used for studies that are the basis of this research. All the human procedures used were in accordance with the ethical standards of the committee responsible for human experimentation (institutional and national), and with the Helsinki Declaration of 1975, as revised in 2013 (http://ethics.iit.edu/ecodes/node/3931).

CONSENT FOR PUBLICATION

As per the guideline, all subjects were informed of the study procedure and their names have been kept confidential.

AVAILABILITY OF DATA AND MATERIALS

The data that support the findings of this study are available on request from the corresponding author.

FUNDING

None.

CONFLICT OF INTEREST

The authors declare no conflict of interest, financial or otherwise.

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SUPPLEMENTARY MATERIAL

Supplementary material is available on the publisher's website along with the published article.

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